Program 4 : Checksum

AIM: To Write a program to implement checksum

Description : A **checksum** is an error-detection technique used to verify the integrity of transmitted data. The sender divides the data into equal-sized words and adds them using **one's complement arithmetic**. The complement of the sum is called the **checksum** and is sent along with the data. At the receiver, the data and checksum are added again. If the result is **all 1s** (or the complement is **0**), the data is considered error-free; otherwise, an error is detected. Checksums are commonly used in **IP, TCP, and UDP** protocols.

Algorithm:

1. Read the number of **8-bit binary words**.
2. Convert each binary word into its decimal equivalent.
3. Add all the values together.
4. If the sum exceeds **8 bits (255)**, perform **end-around carry**.
5. Take the **1's complement** of the final 8-bit sum.
6. Display the checksum in **binary format**.

Program:

```
#include <stdio.h>
#include <string.h>

int binaryToDecimal(char binary[]) {
    int decimal = 0, i;

    for(i = 0; binary[i] != '\0'; i++) {
        decimal = decimal * 2 + (binary[i] - '0');
    }
    return decimal;
}

void decimalToBinary(int num) {
    int i;
    for(i = 7; i >= 0; i--)
        printf("%d", (num >> i) & 1);
}

int main() {
    char binary[10];
    int n, i;
    int sum = 0, checksum;

    printf("Enter the number of 8-bit binary words: ");
    scanf("%d", &n);

    printf("Enter the binary words:\n");
```

```
    for(i = 0; i < n; i++) {
        scanf("%s", binary);
        sum += binaryToDecimal(binary);
    }

    // End-around carry
    while(sum > 255) {
        sum = (sum & 255) + (sum >> 8);
    }

    // 1's complement
    checksum = (~sum) & 255;

    printf("\nFinal Sum      : ");
    decimalToBinary(sum);

    printf("\nChecksum       : ");
    decimalToBinary(checksum);

    printf("\n");

    return 0;
}
```

Output:

```
Enter the number of 8-bit binary words: 3
Enter the binary words:
10101010
11001100
11110000

Final Sum      : 01101000
Checksum       : 10010111
```